



Data Preprocessing



Why Data Preprocessing?

- Data in the real world is soiled.
 - **incomplete**: lacking attribute values, lacking certain attributes of interest, or containing only aggregate data
 - e.g., occupation=""
 - **noisy**: containing errors or outliers
 - e.g., Salary="-10"
 - **inconsistent**: containing discrepancies in codes or names
 - e.g., Age="42" Birthday="03/07/1997"
 - e.g., Was rating "1,2,3", now rating "A, B, C"
 - e.g., discrepancy between duplicate records



What is Data?

- Collection of data objects and their attributes
- An attribute is a property or characteristic of an object
 - Examples: eye color of a person, temperature, etc.
 - Attribute is also known as variable, field, characteristic, or feature
- A collection of attributes describe an object
 - Object is also known as record, point, case, sample, entity, or instance

Attributes				
Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes



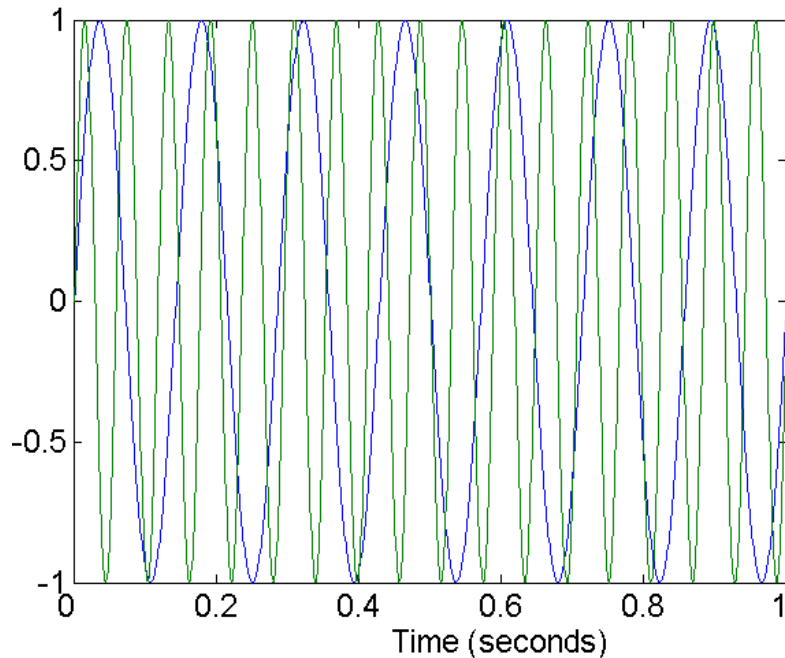
Data Quality

- What kinds of data quality problems?
- How can we detect problems with the data?
- What can we do about these problems?

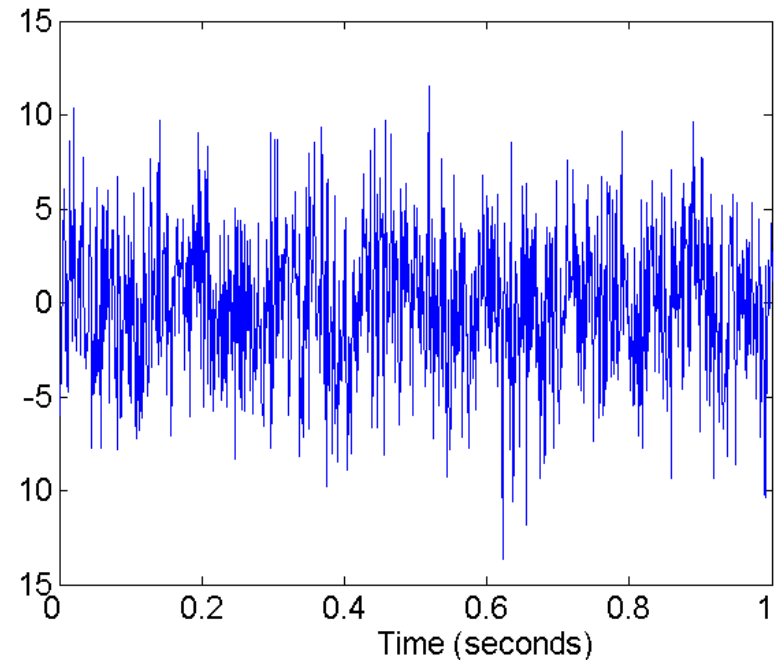
- Examples of data quality problems:
 - Noise and outliers
 - missing values
 - duplicate data



- Noise refers to modification of original values
 - Examples: distortion of a person's voice when talking on a poor phone and “snow” on television screen



Two Sine Waves

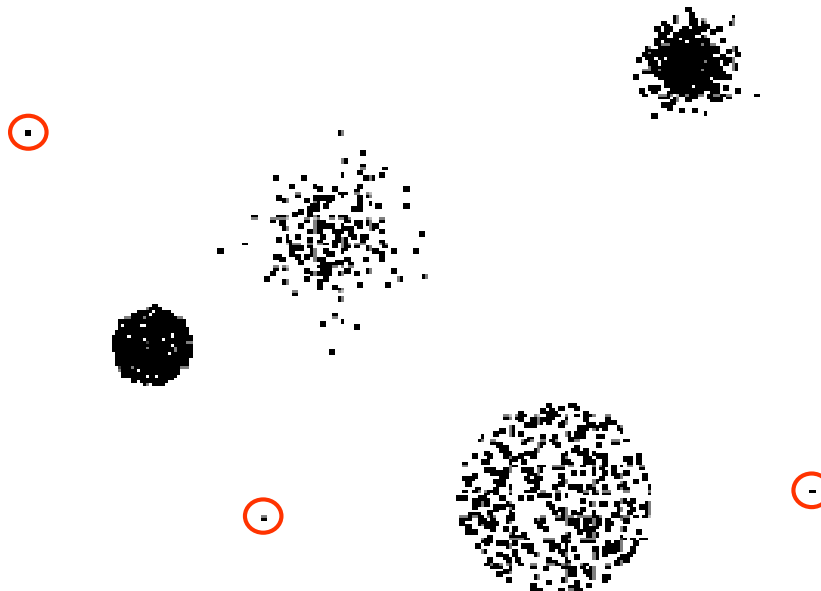


Two Sine Waves + Noise



Outliers

- Outliers are data objects with characteristics that are considerably different than most of the other data objects in the data set





Missing Values

- Reasons for missing values
 - Information is not collected
(e.g., people decline to give their age and weight)
 - Attributes may not be applicable to all cases
(e.g., annual income is not applicable to children)

- Handling missing values
 - Eliminate Data Objects
 - Estimate Missing Values
 - Ignore the Missing Value During Analysis
 - Replace with all possible values (weighted by their probabilities)



Duplicate Data

- Data set may include data objects that are duplicates, or almost duplicates of one another
 - Major issue when merging data from heterogeneous sources
- Examples:
 - Same person with multiple email addresses
- Data cleaning
 - Process of dealing with duplicate data issues



-- ... Why Preprocess the Data

- **Reason for data cleaning**
 - Incomplete data (missing data)
 - Noisy data (contains errors)
 - Inconsistent data (containing discrepancies)
- **Reasons for data integration**
 - Data comes from multiple sources
- **Reason for data transformation**
 - Some data must be transformed to be used for mining
- **Reasons for data reduction**
 - Performance
- **No quality data → no quality mining results!**



Major Tasks in Data Preprocessing

- 1.Data cleaning
 - Fill in missing values, smooth noisy data, identify or remove outliers, and resolve inconsistencies
- 2.Data integration
 - Integration of multiple databases, data cubes, or files
- 3.Data transformation
 - Normalization and aggregation
- 4.Data reduction (Sampling, dimensionality reduction, feature subset selection)
 - Obtains reduced representation in volume but produces the same or similar analytical results



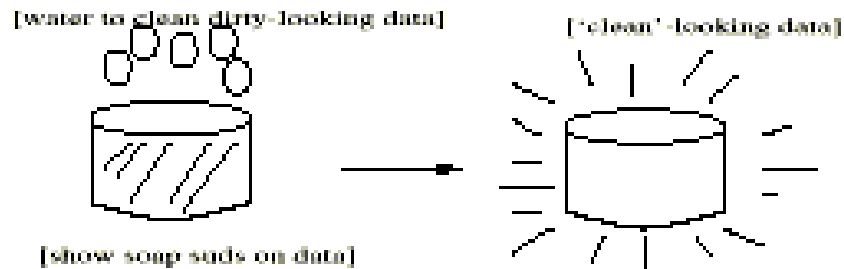
■ 5.Data discretization

- For classification algorithms sometimes it is required that data should be in the form of categorical attributes
- Algo. That find association patterns require that the data be in the form of binary attributes.
- Thus it is required to transform a continuous attribute to a categorical attribute(discretization).
- Part of data reduction but with particular importance, especially for numerical data

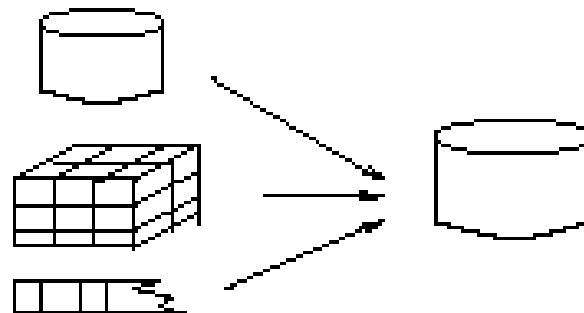


Forms of Data Preprocessing

Data Cleaning



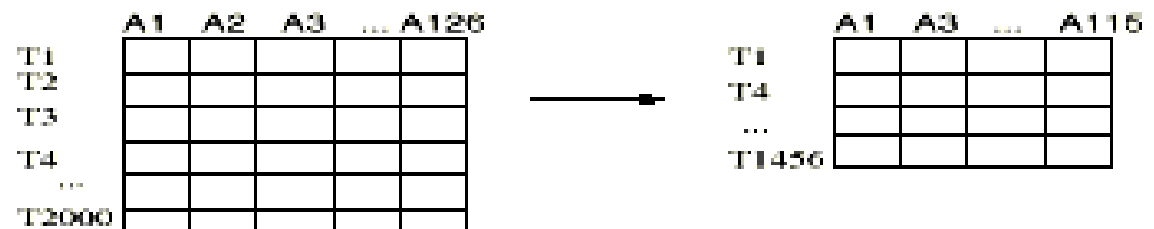
Data Integration



Data Transformation

-2, 32, 100, 59, 48 → -0.02, 0.32, 1.00, 0.59, 0.48

Data Reduction





1.Data Cleaning ●

- Data cleaning tasks
 - Fill in **missing** values
 - Identify outliers and smooth out **noisy** data
 - Correct inconsistent data
 - Resolve redundancy caused by data integration



1.Data Cleaning •

: How to Handle Missing Data? •

- **Ignore the tuple**: usually done when class label is missing (assuming the tasks in classification)—not effective unless the tuple contains several attributes with the missing values
- **Fill in the missing value manually**- not feasible for large datasets and time- consuming
- **Fill in it automatically with**
 - **a global constant** : e.g., “unknown”, a new class?!
 - **the attribute mean**
 - **the most probable value**: inference-based such as Bayesian formula or regression or decision tree induction



1.Data Cleaning : How to Handle Noisy Data? ● ●

- Noise- a random error or variance in a measured variable.
- Incorrect attribute values may due to
 - faulty data collection
 - data entry problems
 - data transmission problems
 - data conversion errors
 - Data decay problems
 - technology limitations, e.g. buffer overflow or field size limits



1.Data Cleaning : How to Handle Noisy Data? •

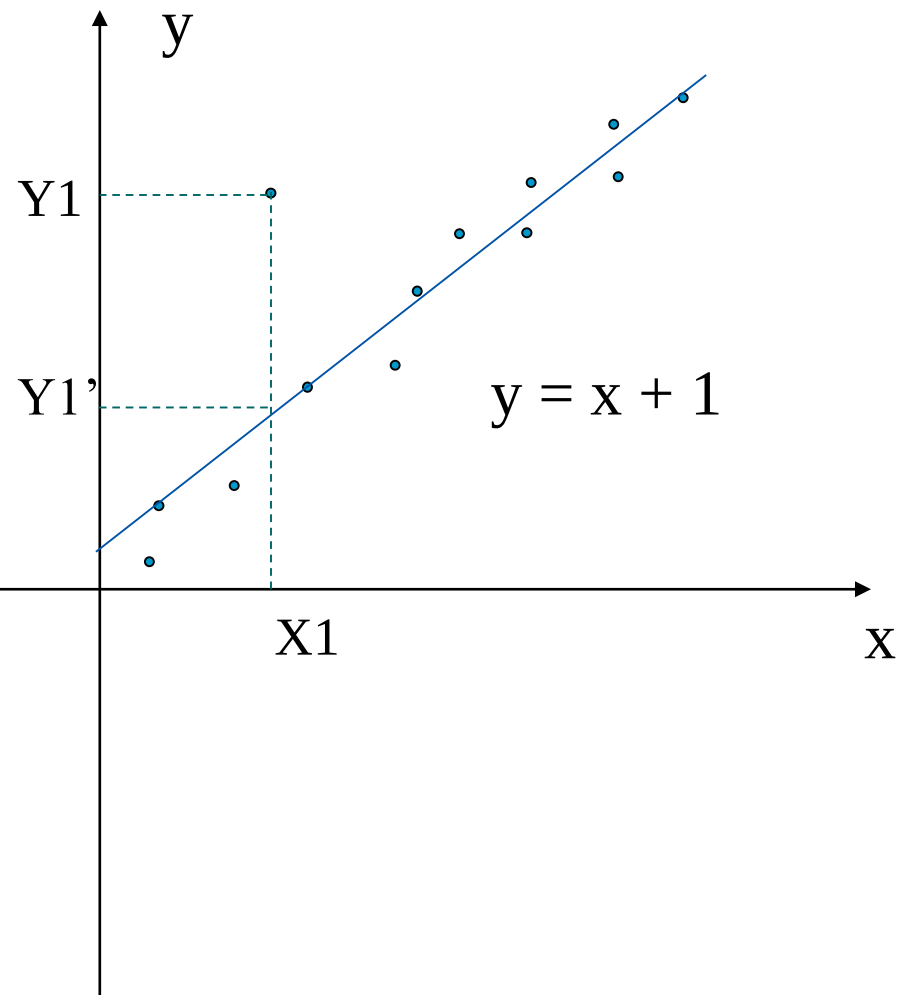
Methods

- **Binning**
 - first sort data and partition into (equal-frequency) bins
 - then one can smooth by bin means, smooth by bin median, smooth by bin boundaries, etc.
- **Regression**
 - smooth by fitting the data into regression functions
- **Clustering**
 - detect and remove outliers
- **Combined computer and human inspection**
 - detect suspicious values and check by human (e.g., deal with possible outliers)



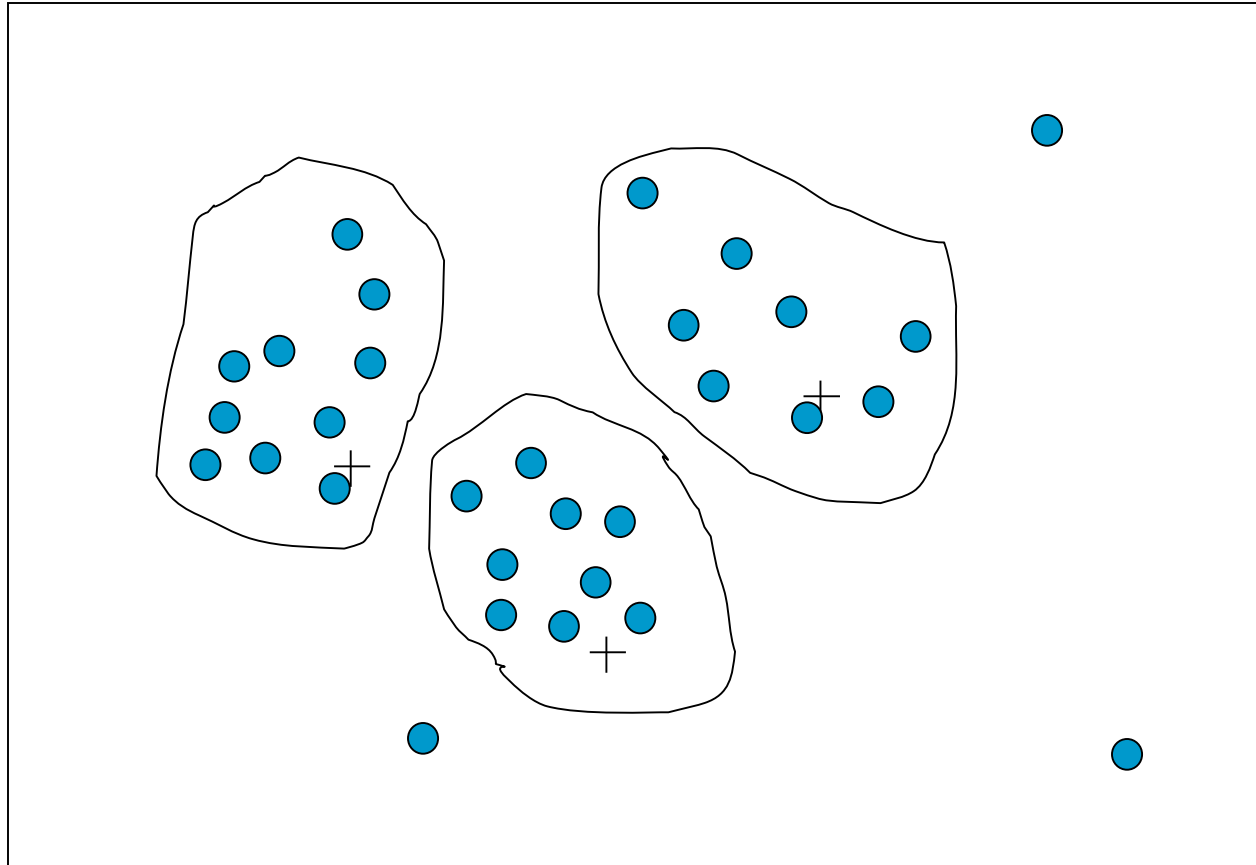
1.Data Cleaning : Regression

- Data can be smoothed by fitting the data to a function such as with regression.
- Linear regression involves finding the 'best' line to fit 2 variables.
- Also, it is possible to predict one variable using the other variable.





1.Data Cleaning : Cluster Analysis





2. Data Integration

- Data integration:
 - Combines data from **multiple sources into a coherent store**
- Schema integration: e.g., $A.cust-id \equiv B.cust-\#$
 - Integrate metadata from different sources
- **Entity identification problem:**
 - Identify real world entities from multiple data sources, e.g., Bill Clinton = William Clinton
- Detecting and resolving data value conflicts
 - For the same real world entity, attribute values from different sources are different
 - Possible reasons: different representations, different scales



Data Integration •

: Handling Redundancy in Data Integration •

- Redundant data occur often when integration of multiple databases
 - *Object identification*: The same attribute or object may have different names in different databases
 - *Derivable data*: One attribute may be a “derived” attribute in another table, e.g., annual revenue
- Redundant attributes may be able to be detected by *correlation analysis*
- Careful integration of the data from multiple sources may help reduce/avoid redundancies and inconsistencies and improve mining speed and quality



Data Transformation

- Smoothing: remove noise from data
- Aggregation: summarization, data cube construction
- Generalization: concept hierarchy climbing
- Normalization: scaled to fall within a small, specified range
 - min-max normalization
 - z-score normalization
 - normalization by decimal scaling
- Attribute/feature construction
 - New attributes constructed from the given ones



Data Reduction Strategies

- Why data reduction?
 - A database/data warehouse may store terabytes of data
 - Complex data analysis/mining may take a very long time to run on the complete data set
- Data reduction
 - Obtain a reduced representation of the data set that is much smaller in volume but yet produce the same (or almost the same) analytical results
- Data reduction strategies
 - Aggregation
 - Sampling
 - Dimensionality Reduction
 - Feature subset selection
 - Feature creation
 - Discretization and Binarization
 - Attribute Transformation



Data Reduction : Aggregation

- Combining two or more attributes (or objects) into a single attribute (or object)
- Purpose
 - Data reduction
 - Reduce the number of attributes or objects
 - Change of scale
 - Cities aggregated into regions, states, countries, etc
 - More “stable” data
 - Aggregated data tends to have less variability



Data Reduction : Sampling

- Sampling is the main technique employed for data selection.
 - It is often used for both the preliminary investigation of the data and the final data analysis.
- Statisticians sample because **obtaining** the entire set of data of interest is too expensive or time consuming.
- Sampling is used in data mining because **processing** the entire set of data of interest is too expensive or time consuming.



Thank You